DECENCY, REASONABLENESS AND PROPERTY (BETA) OF THE SOURCE TRANSFER TO THE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3943-3950, immediately following the statement at 3935-3941.

Editorial note. Two same-chapter line-indexed stubs are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Representable morphisms inherit the named properties from a decent or reasonable source. Lemma lemma-representable-named-properties of this chapter, cited at proof lines 3947-3948. T x_Y X -> X representable, and T x_Y X is decent/reasonable when X is – carries the decent and reasonable cases.
- Representable morphisms and property (beta). Lemma lemma-representable-properties of this chapter, cited at proof lines 3949-3950. Carries the property (beta) case separately.

Source reference. Original lemma 03JX: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 952–952. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ABT: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1421–1421. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03KF: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1126–1126. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If X is decent (resp. is reasonable, resp. has property (β) of Lemma 03JX), then f is decent (resp. reasonable, resp. has property (β)).*

Proof. Let T be a scheme and let $T \to Y$ be a morphism. Then $T \to Y$ is representable, hence the base change $T \times_Y X \to X$ is representable. Hence if X is decent (or reasonable), then so is $T \times_Y X$, see Lemma 0ABT. Similarly, for property (β), see Lemma 03KF. □

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